

Foundation (Jalapeno)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^{3+4}
- (B) $2^{3 \cdot 4}$
- (C) 2^{3-4}
- (D) $2^{3/4}$
- (E) 2^{4-3}

Q2. A basketball team scores 3^2 points in the first quarter and 3^3 points in the second quarter. What is the total number of points scored?

- (A) 3^{2+3}
- (B) $3^{2 \cdot 3}$
- (C) 18
- (D) 27
- (E) 12

Q3. Simplify $(x^2)^3$

- (A) x^{2+3}
- (B) $x^{2 \cdot 3}$
- (C) x^{2-3}
- (D) $x^{2/3}$
- (E) x^{3-2}

Q4. A cyclist travels 2^4 miles in the first hour and 2^5 miles in the second hour. What is the total distance traveled?

- (A) 2^{4+5}
- (B) $2^{4 \cdot 5}$
- (C) 40
- (D) 32
- (E) 48

Q5. Simplify

$a^5 a^3$

- (A) a^{5+3}
- (B) a^{5-3}
- (C) $a^{5 \cdot 3}$
- (D) $a^{5/3}$
- (E) a^{3-5}

Q6. A runner's time is 2^2 minutes for the first lap and 2^3 minutes for the second lap. What is the total time taken?

- (A) 2^{2+3}
- (B) $2^{2 \cdot 3}$
- (C) 12
- (D) 10
- (E) 8

Q7. Simplify $x^3 \cdot x^2$

- (A) x^{3+2}
- (B) x^{3-2}
- (C) $x^{3 \cdot 2}$
- (D) $x^{3/2}$
- (E) x^{2-3}

Q8. A tennis player hits 3^2 aces in the first set and 3^3 aces in the second set. What is the total number of aces hit?

- (A) 3^{2+3}
- (B) $3^{2 \cdot 3}$
- (C) 18
- (D) 30
- (E) 12

Open Ended Questions (FRQ)

Q9. Simplify $(2x^2)^3$ and show all steps

Q10. A golfer's score is x^2 strokes for the first round and x^3 strokes for the second round. Simplify the total score $x^2 + x^3$ and show all steps

Chef's Tips

- Chef's Tips
- Check units and dimensions
- Make sure to simplify expressions